The Google PageRank Score

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December 5, 2024

Acknowledgments

- I would like to acknowledge and thank Professor Jessica Lin and fellow researcher Yifu Zhang for their guidance and collaboration throughout this course.
- I would also like to say a special thank you to Professor Louigi Addario-Berry for co-supervising this report.

Definitions and Notation

- A **directed graph** G has $n \in \mathbb{N}$ vertices such that each edge has a direction from an origin to a target.
- A **web graph** $G = (V, E)$ is a finite directed graph such that V is the set of pages and E is the set of hyperlinks.
- An **adjacency matrix** A is a matrix that shows which vertices are connected by an edge in the graph, where the entry has a 1 if there is a connection and 0 otherwise.
- An **eigenvalue** λ is a non-zero real number that satisfies the equation $Mv = \lambda v$, for a matrix M and vertex v.
- The **out-degree** of a node $deg_{out}(v_i)$ is the number of nodes that have v_i as the origin.

Why PageRank?

- The internet can be viewed as a massive network (directed graph) of interconnected pages (vertices).
- **Goal:** Identify which pages are the most "important" based on their connections.
- **Challenge:** Not all pages have an equal importance; it is based on what other pages are linking to it.
- **Solution:** The PageRank score and algorithm, developed by Google in 1998 [BP98].

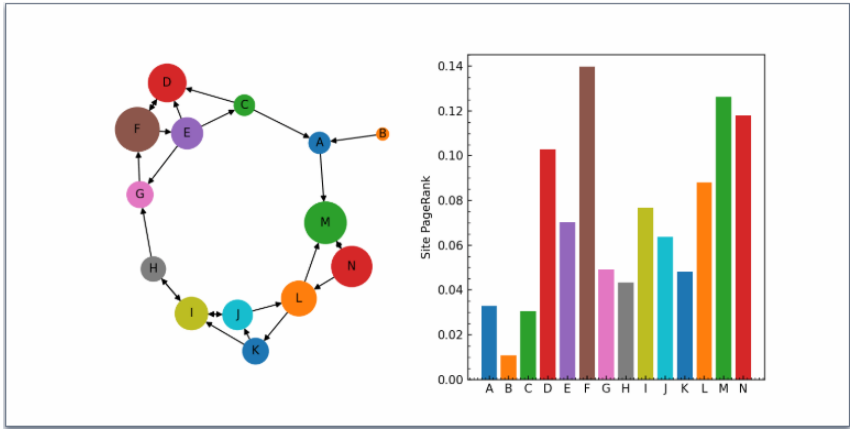

Figure: PageRank in Action [Wik24]

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Centrality Vector

For a graph $G = (V, E)$ with n nodes, the centrality vector q is:

$$q = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{pmatrix}$$

where q_i is the importance score of node v_i.

The Original PageRank Score (1998)

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Let $G = (V, E)$ represent the web graph. The **PageRank score** for page v_i is given by:

$$r_i = \sum_{j=1}^{n} \frac{A_{ji}}{deg_{out}(v_j)} r_j \tag{2.1}$$

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Issues with Original Model:

- Dangling nodes (pages with no out-going links)
- Rank sinks (importance accumulates infinitely in cycles)

Katz-Bonacich Centrality Score

Katz-Bonacich Centrality of a Node

Let A be the adjacency matrix of a graph and λ the largest eigenvalue. We compute the centrality of a node v_i by:

$$x_i = \frac{1}{\lambda} \sum_{j=1}^{n} A_{ji} x_j + \beta \tag{2.2}$$

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Role of λ: It controls how much influence a node receives from its neighbors.
Role of β: This term weighs the contribution of paths based on their lengths.

Katz-Bonacich PageRank Modification

PageRank Score of a Node

Let G be our web graph, A its adjacency matrix, $\alpha \in [0, 1)$. The *PageRank score* of a node is:

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Iterative Process:

- Start with $r^{(0)} = \mathbf{1}$ (equal importance for all nodes) for time-step k=0.
- Update at each step: $r^{(k+1)} = \alpha H r^{(k)} + \mathbf{1}$
- Repeat until $||r^{(k+1)} - r^{(k)}|| < \epsilon$.

where the hyperlink matrix H is a *modified transition matrix* [Mun17], [Wol22].

The Parameters: Usage and Limitations

We will now analyze the three parameters $(H, \alpha, \beta = \mathbf{1})$ in the vector r of PageRank scores. Recall:

$$r^{(k+1)} = \alpha H r^{(k)} + \mathbf{1}$$

The Hyperlink Matrix H:

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The Hyperlink Matrix H: We described the hyperlink matrix H as a *modified transition matrix*. We define this as:

$$H_{ij} = \begin{cases} \frac{A_{ji}}{deg_{out}(v_j)} & \text{if } deg_{out}(v_j) \neq 0 \\ 1 & \text{if } deg_{out}(v_j) = 0 \end{cases}$$

Importance of $deg_{out}(v_j)$?

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- Human behavioral patterns
- Lay-out of webpage
- Webpage topic versus the hyperlink
- Many, many more choices!

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The β=1 Parameter:

- We have that $\beta = 1$. This assumes that all paths in the network, regardless of their length, contribute equally to the importance of the node.
- This ensures that dangling nodes with no out-degree have a "standardized" importance rating, rather than it be set to zero (which would skew the overall ranking results).

Limitations of Katz-Bonacich PageRank

- This model only works assuming there are **no cycles** in the network!
- If there are cycles, we see something called a **rank sink** (which is a dangling-node that has a cycle pointing to it).
- **Why is this bad?** The node will have infinitely increasing score each iteration!

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- Recall $\beta = 1$ means that all paths have equal contribution to the algorithm.
- But this means that much longer paths have equal influence to shorter paths.
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What can we do to fix this? Markov Chains!

Markov Chains and Convergence

Markov Chains Definitions:

- A **stochastic process** is a set of random surfers paired with a state space of pages, with time dependency.
- A **Markov Chain** is a stochastic process such that future states of the chain depend only on the current state, not on the past history of the chain.
- A **stochastic matrix** S is such that each column sums to 1.
- A stochastic matrix is **irreducible** if it represents a Markov chain in which every state is reachable from every other state.

The Google Matrix

Construction and Definition: $G = \alpha S + (1 - \alpha)E$

- S: A stochastic transition matrix (slightly modified version of Hyperlink Matrix).
- E: A teleportation matrix (you can now exit the tab and go to a new one).
- $\alpha = 0.85$ as before.

Features of Google Matrix

$$G = \alpha S + (1 - \alpha)E \tag{4.1}$$

Key Features:

- **Stochastic and Irreducible**: Every node contributes to rankings, including dangling-nodes (S parameter).
- **Random teleportation:** The E parameter prevents rank-sinks; this helps redistribute scores globally.

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By Perron-Frobenius Theorem, G has:

1. Iterative computation $r^{(k+1)} = r^{(k)}G$ converges to a unique vector.
2. This unique stationary vector is the PageRank Vector!

Any Questions?

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Google Matrix Definitions

We define the modified Hyperlink Matrix S as:

$$S = H + \delta_d(\frac{1}{n}\mathbf{e}^T)$$

where δ_d is a vector such that

$$\delta_i = \begin{cases} 1, & \text{if } v_i \text{ is a dangling node} \\ 0, & \text{otherwise} \end{cases}$$

We define the Teleportation Matrix E as:

$$E = \frac{1}{n}\mathbf{e}\mathbf{e}^T$$

We have that teleporting within a network is random.